

SAN BERNARDINO, Switzerland

WMO: 067830

Lat: 46.4636N

Lon: 9.1847E

Elev: 5374

StdP: 12.06

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.5	10.7	-10.1	3.9	17.2	-5.7	4.9	17.9	26.1	21.4	23.3	22.9	5.8	320	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	71.5	57.2	68.8	56.6	66.2	55.4	60.3	67.2	58.6	65.4	57.0	63.5	6.2	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.6	86.6	63.6	55.9	81.4	61.6	54.3	76.8	60.1	29.3	67.3	28.0	65.8	26.8	63.6	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	18.0	15.8	2.5	75.6	4.1	3.4	-0.4	78.1	-2.8	80.1	-5.1	82.0	-8.1	84.4		
(5)			1.2	62.9	4.0	1.9	-1.7	64.3	-4.1	65.4	-6.3	66.5	-9.2	67.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	25.7	26.4	31.7	37.2	44.9	52.8	56.0	55.4	48.5	42.1	33.2	27.8
	DBStd	12.51	6.81	7.64	6.89	5.91	5.61	5.93	4.78	4.87	5.31	6.07	6.89	6.97
	HDD50	4132	753	661	569	385	176	38	10	10	88	251	503	688
	HDD65	9047	1218	1081	1034	835	623	367	279	300	494	710	953	1153
	CDD50	566	0	0	0	1	18	122	197	177	44	6	1	0
	CDD65	3	0	0	0	0	0	1	0	2	0	0	0	0
	CDH74	24	0	0	0	0	0	8	3	13	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	5.4	6.3	6.9	6.4	5.5	5.1	5.3	5.4	4.6	4.7	4.3	5.0	5.5
Precipitation	PrecAvg	60.4	2.7	2.2	2.7	4.9	6.3	6.6	6.2	6.7	6.3	6.2	6.7	2.9
	PrecMax	84.7	7.9	8.3	9.3	13.8	18.1	15.5	11.1	11.5	14.8	20.5	27.4	7.6
	PrecMin	33.7	0.3	0.3	0.3	0.9	1.7	1.5	2.0	1.1	1.2	0.7	0.5	0.1
	PrecStd	12.8	2.0	1.8	2.0	3.1	3.5	3.3	2.4	2.3	4.3	4.2	6.1	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.5	50.0	52.3	58.9	66.4	74.2	73.8	75.6	67.9	62.5	55.6	48.5
		MCWB	35.6	36.5	38.6	45.2	52.3	57.6	58.8	58.1	53.5	47.9	42.0	35.1
	2%	DB	42.0	44.8	48.7	54.0	62.2	70.4	70.9	71.2	64.1	57.7	49.4	43.3
		MCWB	32.8	34.1	37.3	42.2	49.3	56.7	57.4	57.6	52.9	45.8	39.9	33.4
	5%	DB	38.7	41.0	45.2	50.2	58.6	67.1	68.4	67.7	60.8	54.4	45.2	39.8
		MCWB	30.9	32.2	35.7	40.5	47.8	55.4	56.9	56.6	51.6	45.0	37.5	32.0
	10%	DB	35.0	37.1	41.9	46.5	54.8	64.0	65.8	64.7	57.6	51.1	41.9	36.8
		MCWB	29.3	29.7	34.0	38.5	46.3	54.0	55.8	55.6	50.3	44.5	36.9	30.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	37.8	40.5	46.6	54.0	61.9	63.0	62.1	57.4	51.0	44.7	38.0
		MCDB	45.4	46.8	48.7	56.1	62.5	68.8	69.6	69.4	63.4	56.9	51.5	43.7
	2%	WB	34.4	34.8	38.4	43.8	51.4	59.0	60.5	59.9	55.1	49.2	41.8	35.6
		MCDB	39.5	42.9	45.2	51.3	59.0	66.8	67.0	66.9	60.7	53.1	46.3	39.7
	5%	WB	32.3	32.7	36.7	41.6	49.2	56.8	58.7	58.2	53.1	47.6	39.9	33.6
		MCDB	35.9	39.1	43.1	47.8	56.1	64.5	65.1	64.8	58.3	51.2	43.1	37.3
	10%	WB	30.5	30.9	35.0	39.6	47.3	54.9	57.1	56.6	51.4	46.1	38.0	32.1
		MCDB	34.2	35.9	41.0	44.9	53.0	62.3	63.4	62.5	55.9	49.6	40.9	35.4
Mean Daily Temperature Range	5% DB	MDBR	12.5	12.8	12.9	12.7	13.8	15.2	15.4	14.7	13.7	13.0	10.8	11.9
		MCDBR	17.3	17.2	17.5	19.1	20.1	20.3	19.4	19.5	18.9	19.4	17.4	16.4
		MCWBR	11.0	10.4	9.9	10.3	11.1	11.0	10.5	10.6	10.6	11.0	10.5	10.2
	5% WB	MCDBR	13.6	15.1	14.9	16.5	17.7	18.6	18.1	17.6	16.1	14.3	13.2	12.5
		MCWBR	9.6	9.9	9.1	9.9	10.9	11.3	11.1	10.7	10.1	9.3	9.0	8.9
Clear-Sky Solar Irradiance	taub		0.240	0.264	0.307	0.353	0.367	0.377	0.370	0.362	0.342	0.316	0.272	0.237
	taud		2.332	2.268	2.203	2.184	2.280	2.300	2.319	2.358	2.399	2.448	2.445	2.368
	Ebn at Noon		278	290	291	285	285	282	282	280	276	268	263	267
	Edh at Noon		23	30	38	43	41	40	39	36	31	26	21	20
All-Sky Solar Radiation	RadAvg		510	818	1204	1418	1521	1790	1791	1552	1209	801	497	426
	RadStd		64	95	174	229	151	157	154	136	117	95	75	55

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.70	N/A	N/A	+1.30	+1.01	+0.76	N/A	-272	N/A	N/A
(47) Regional (117 neighbors)		N/A	N/A	N/A	+1.30	+0.76	+0.58	N/A	N/A	+108	+31

Nomenclature: See separate page